

PSI 1601 Analog Fiber-Optic Link

Technical Specifications
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Photonic Systems, Inc.
900 Middlesex Turnpike
Building #5
Billerica, MA 01821

Phone: (978) 670-4990
FAX: (978) 670-2510
email: info@photonicsinc.com

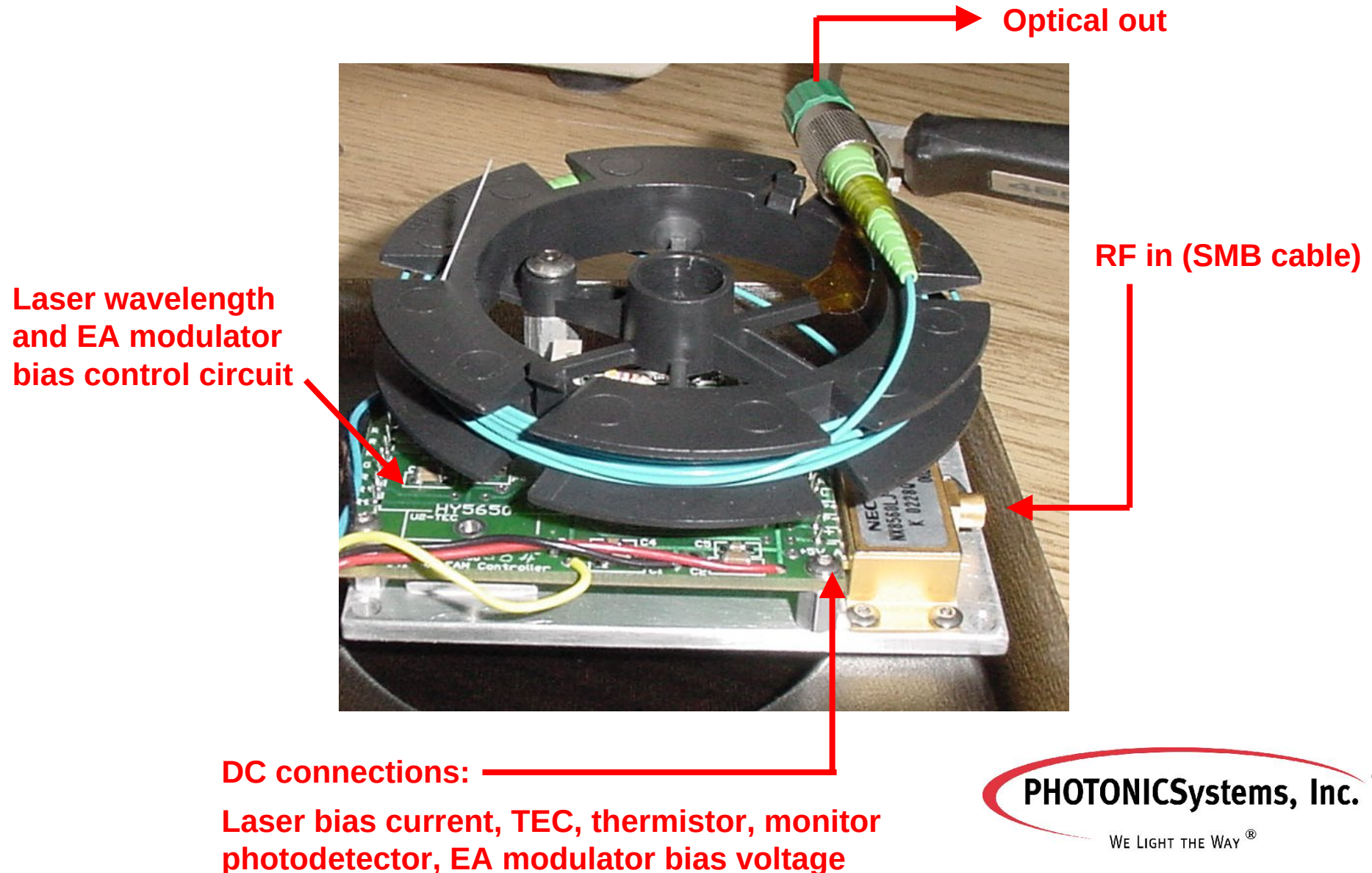


PSI 1601 Analog Fiber-Optic Link

- Link consists of optical transmitter and optical receiver modules
- Photographs on following two pages are of the minimally packaged (i.e., printed circuit board) version; custom package styles available on request
- Interfaces:
 - Optical interfaces to transmitter, receiver modules are short (< 1-meter) single-mode fiber pigtails
 - RF interfaces are SMA connectors

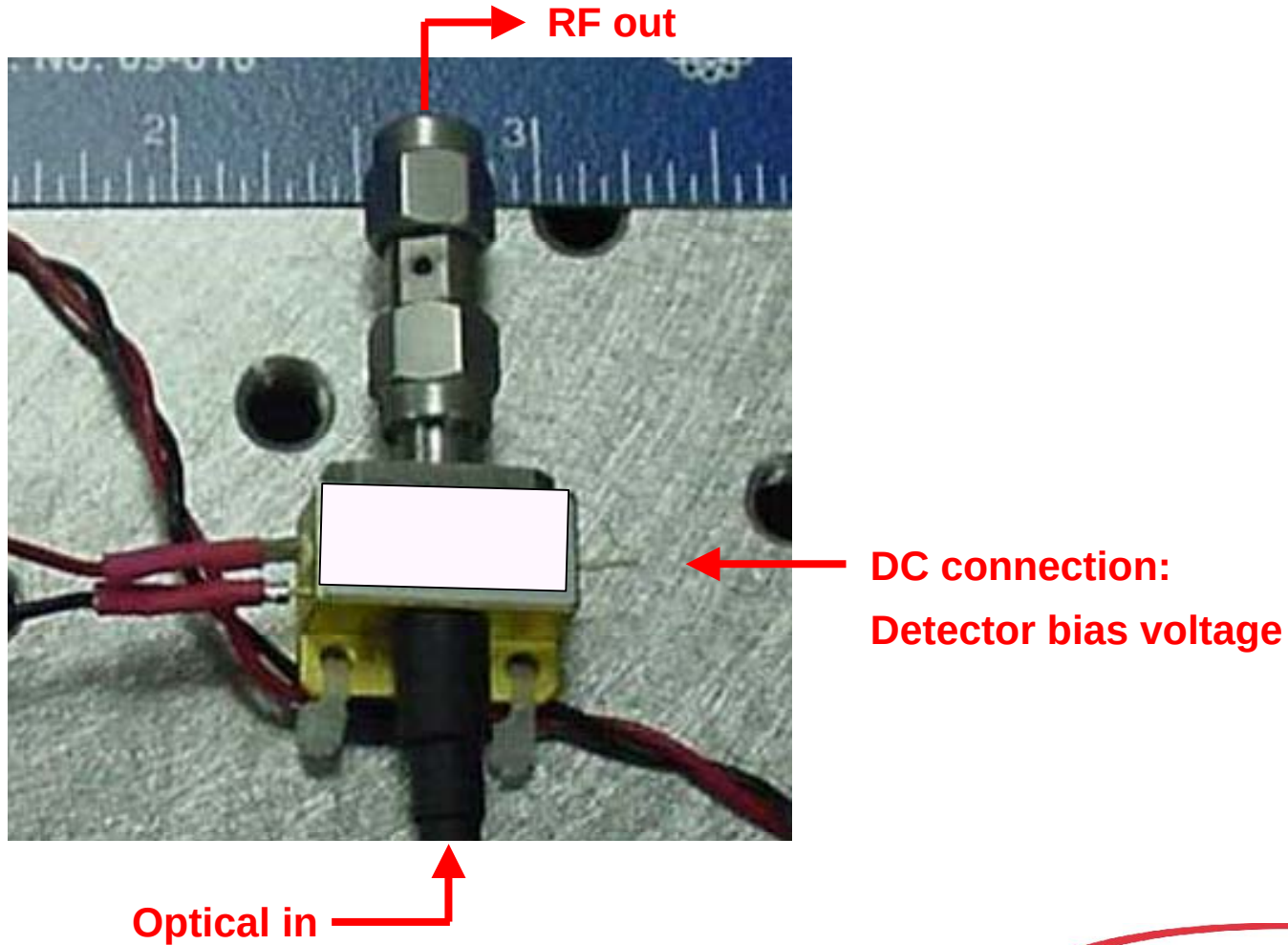
Optical Transmitter Module

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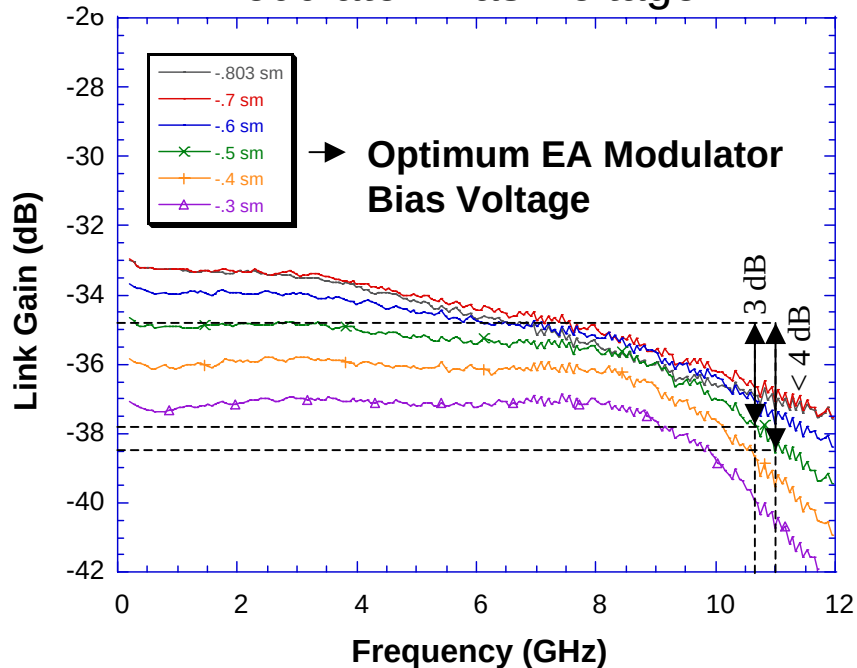
Optical Receiver Module

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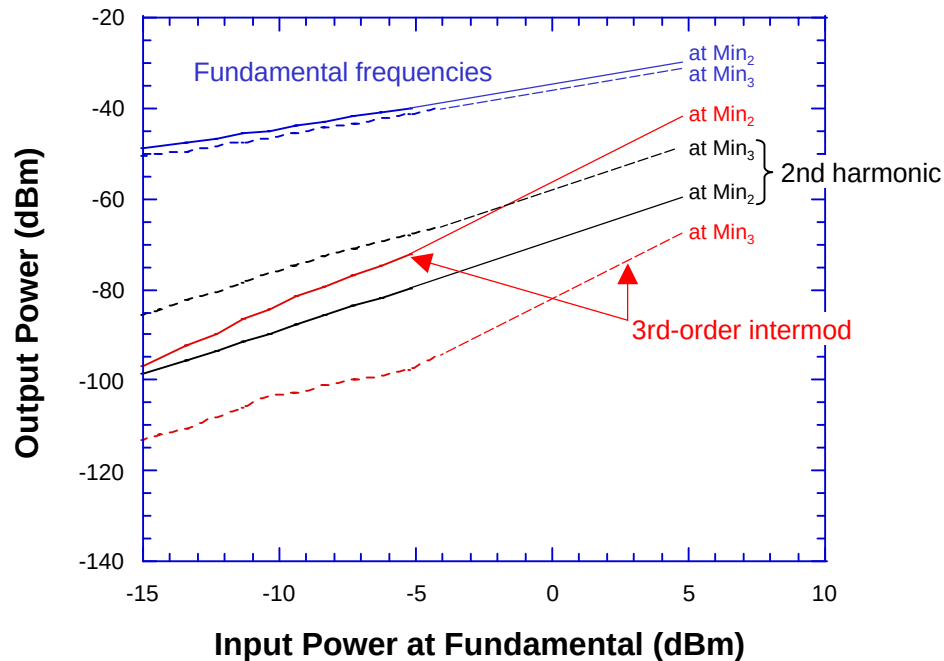


Measured PSI 1601 Analog Fiber-Optic Link Performance vs. Frequency (Typical)

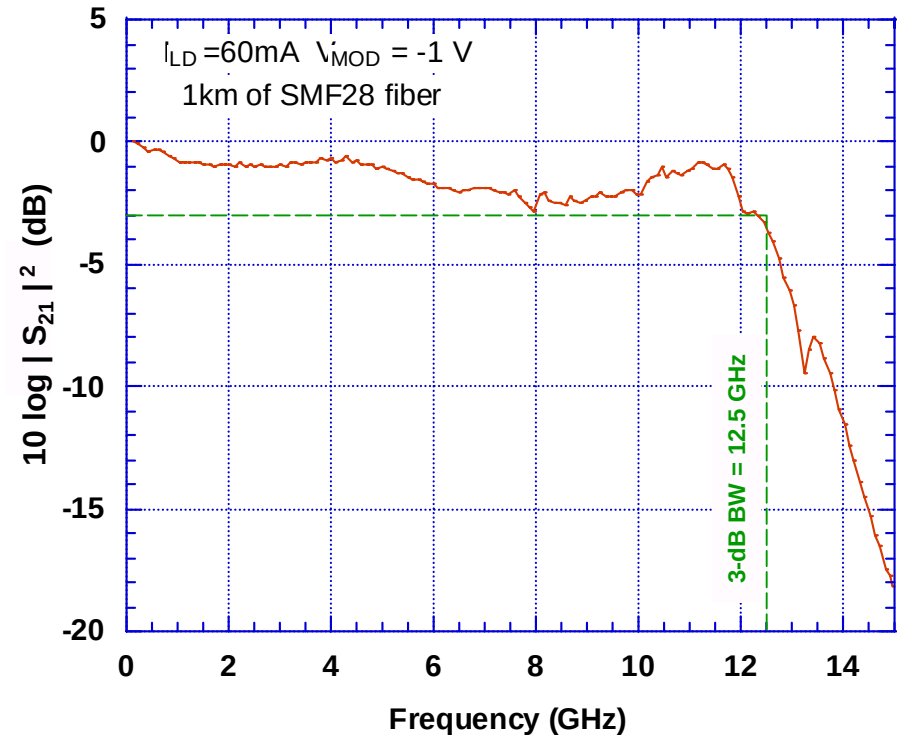
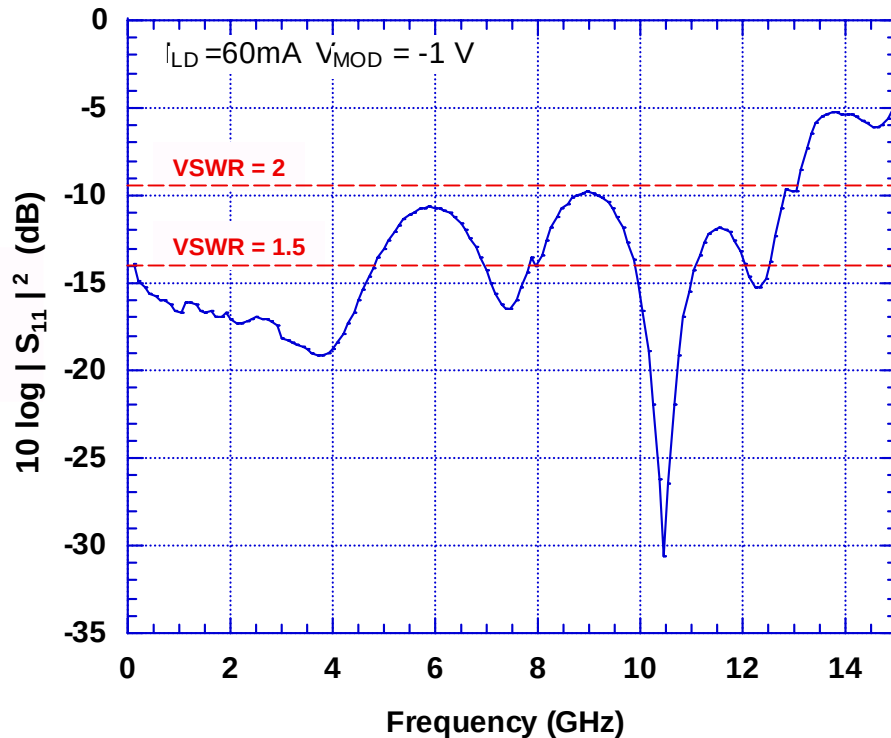
Measured Link Frequency Response vs. Modulator Bias Voltage



Measured Distortion Products
(detector bias = 6 V, fundamental frequencies ~ 5 GHz)



S_{11} and S_{21} (VSWR and Frequency Response) PSI 1601 Analog Fiber-Optic Link



Summary of Typical PSI 1601 Analog Fiber-Optic Link Performance

Parameter	Value
3-dB Bandwidth	0.5 – 12 GHz
Gain	≥ -40 dB
VSWR	$\leq 2:1$ across BW
Noise Figure	≤ 45 dB
Third-order intermodulation-free dynamic range	≥ 100 dB in 1 Hz
1-dB compression dynamic range	≥ 130 dB in 1 Hz
Maximum fiber length *	≥ 1 km
Fiber-optic connectors	FC/APC **
DC power requirement	≤ 7.5 W (transmitter), ≤ 25 mW (receiver)

* Greater fiber lengths are possible with reduced performance;
link does not include 1 km of fiber

** Standard; other connector types are available